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The Semiotics of Time Structure in Ludic Space As a Foundation for Analysis and Design¹

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Abstract

The concept of a ludic systems encompasses a family of media forms and experiences involving elements of simulation, game play and narrative or story construction. These three elements can be regarded as different classes of semiotic systems, or systems of meaning, having their own structuring principles and methods of informing experience. For any particular ludic system, such as a computer game, time structure can be considered in terms of a number of distinct layers of meaning analogous to the levels of encoding identified in structuralist narrative theory: a generation level, a simulation level, a performance level and a discourse level. The simulation, performance and discourse levels correspond with the semiotic domains of simulations, games and narratives. For any specific ludic system, the overall design approach relating to how the designer intends the players' experience to be structured, as the core of interactive engagement and immersion, can be based upon emphasizing one of these three primary forms, or integrating more than one form by various strategies. Adopting a structural semiotic approach to modeling these layers of meaning provides a foundation for more clearly integrating design choices within a coherent overall concept, as well as laying the foundations for a more systematic study of possible correlations between design features and player affects.

Introduction

The concept of 'ludic space' captures systems of experience incorporating concepts of game or game play and related experiences. This is rather broad, but the term captures many forms within the current scope of interest of academic ludology. The forms of interest include computer games that may range from computational versions of simple board games such as chess or checkers, through more complex interactive 3D productions like actions games, role-playing games, strategy games, etc., to flight simulators and even interactive movies having a hypertext structure connecting linear cinematic sequences (e.g. Brilliant Digital's multi-path movies; <http://www.brilliantdigital.com/solutions/movies/>). Ludic space also includes highly diverse non-computational game forms, including simple table-top games, live-action and table-top role-playing games, reality games and sports games. Hybrid forms include augmented-reality games and pervasive games (e.g. www.pervasive-gaming.org). This is a very diverse range of forms, many of which only marginally seem to fit the term 'games', but the extremes can be regarded as limited cases in a broad field within which many design issues are globally relevant. These forms are here collectively referred to as 'ludic systems', and the imaginative space of all possible ludic systems is referred to as 'ludic space', in order to capture this diversity within which 'a game', 'a simulation' or 'a narrative' nevertheless has its own particular design principles and modes of creating meaning.

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The evolution of game design as a discipline requires a systematic approach to understanding ludic systems in all of their diversity. In particular, it is necessary to establish principled decompositions of the elements of ludic systems, to identify significant design options, to understand the interaction modes facilitated by these design options, and to understand potential correlations between game elements, design features, interaction patterns, and affects upon players in relation to their motivations, pleasures and play style preferences. Player affects may be emotional, rhetorical, cognitive, performative, etc..

The aim of seeking a framework within which design features and strategies can be correlated with player affects is ambitious, but represents a broad program to evolve game design into a principled and autonomous creative discipline analogous to established design disciplines such as industrial design or architecture. The overall program can be viewed as a program to develop a semiotics of ludic systems as a foundation for analysis and design. Semiotics may be broadly understood as the investigation and development of theories and models of the creation of meaning. Ludic semiotics begins with a broad conception of the field of ludic systems, and then seeks a systematic framework for the analysis of the ways in which meanings are generated for the players of ludic systems. A computational ludic semiotics approaches this task from the perspective of developing models and languages supporting more coherent and effective ways of specifying, designing and implementing ludic systems, or components of ludic systems, based upon computational technology. In a mature form this is not just a matter of accounting for how to design for specific affects, but should be an ongoing and open process, facilitating innovation as well as operating within established design conventions. The investigation of a specifically semiotic approach is inspired by semiotic analyses in other domains and aims to use principles from semiotic theory that may be applied to game development, semiotics being a science of the microstructures of meaning that may hold the potential to develop design principles based upon subtle differences, as well as the kinds of large scale differences addressed in this kind of overview paper. Adopting a semiotic approach motivated by the development of a design science for computer games in particular leads to an emphasis upon formalist and structuralist methods, since structure is so fundamental to software development. Hence the potential outcomes of a semiotically inspired game design method are not only relevant to game design but can also provide foundations for more advanced implementation strategies, for instance, based upon more declarative and portable representations of game elements and logic. A combat system, or a story template, for example, might be used without modification for a table-top role-playing game, for any number of computer games, for text-based adventure games, or for technologically enhanced live-action role-playing games; one game system or structure model might be used across genres, media and game staging strategies, if it is represented at an appropriate level of abstraction, and if there are well developed methods for deploying it and integrating it within an overall architecture for a specific game.

This paper focuses upon developing a high level view of the time structure of ludic systems from a structural perspective, drawing from semiotic theory and proposing a layered model of time structure within which each layer is its own semiotic domain. Being an autonomous semiotic domain means having a different tradition, language, variety of methods, problems and approaches to developing solutions. This high level perspective of ludic systems design provides a kind of meta-level semiotic domain within which each of these narrower domains have a place and that contains the discourse describing their relationships and providing explicit foundations for designing their integration. The aim is not to duplicate existing material of the kind commonly found in game design references (e.g. Rollings and Adams,

2003, Björk and Holopainen, 2005). Rather, the aim is to define a high level structural framework as part of a process of typologically separating design concerns, features, strategies, principles and patterns based upon a broad conception of their semiotic function and as a prelude to and foundation for a more systematic study of the relationships between design solutions and player affects.

Layers of Encoding Within Ludic Systems

Ludic systems are fundamentally time-based, and temporal structure is a major determinant of the player's perception and experience of ludic form. As with other temporal experiences, the time structure of ludic systems can be regarded in terms of several layers of the encoding of meaning, each being focused upon a different time scale. These layers are hierarchical (unlike those described by Metz, 1974, for cinema, or by Kress and Van Leeuwen, 2001, for multimodal discourse), deriving from Saussure's *langue/parole* distinction (language versus speech), and from the Russian formalist distinction between *fabula* and *syuzhet* (roughly equivalent to story and plot, respectively) in the study of narrative (see Stam et al, 1992).

Ludic systems involve four levels of temporal structure. The temporal structure of the experience of the player, which can be referred to as the *discourse* level, corresponds in verbal, textual and cinematic narrative systems with the level of narration (narrative terms used in this section are dealt with at length in Chatman, 1978, Rimmon-Kenan, 1983 and Stam et al., 1992). This is the level at which a plot is revealed (i.e. represented) via one or more discursive episodes (e.g. a written novel, or different game sessions in the case of a game), and for which the sequence of revealing episodes within the players' temporal continuum may not necessarily correspond with the sequence of temporal events within the revealed plot; that is, the order of revelation of ludic world events to the player does not necessarily reflect the linear time order of that world, but could include flash-forwards and flash backs.

The actual events revealed to the player as part of the play experience may be referred to as the *performance* level. This is the level at which the player is not simply an active viewer, but an interactive participant within the ludic world, having an influence on the nature and shape of the events manifested within a game world during the playing of a game. The performance level includes only those parts of the virtual world directly experienced by the player. In purely narrative systems this is the *plot*. Ludic systems may not have a sufficiently strongly pre-specified plot structure to represent progress within a strongly preconceived (i.e. pre-authored) conception of a narrative, so the performance level cannot generally be regarded as a plot as it is experienced by the player. It may be the case that the play experience is designed to lead to inevitable plot points, either enacted by the player or revealed by non-interactive animation sequences or cut scenes. But interactivity in determining the detail of the plot (and the underlying story), at least between cut scenes, changes the nature of this level in comparison with traditional linear narratives.

The nature of the performance level can be illustrated by considering the example of *The Elder Scrolls III: Morrowind* (Bethesda Softworks, 2002), perhaps one of the most successful first-person computer role playing games in terms of compelling storylines build into a highly non-linear world. The player may choose whether or not to follow the main plot of becoming the saint and savior Nerevar and defeat the game's main villain Dagoth Ur. There is a very specific set of central plot points within this main plot. But the plot points are partially ordered: seven high level tasks must be completed, but their constituent sub-tasks, such as

being named Nerevarine by the four Ashlander tribes, can be accomplished in any order, and this is repeated for the sub-tasks involved in those sub-tasks. The final order is therefore largely chosen by the player, and between the plot points of the central story there is a very broad scope for distraction into many other quests and open, player-defined tasks. Choosing to follow the central plot is very much like choosing a part in a drama, but with a large scope for character interpretation including endless variations of detail along the way, using the language of moves and scenarios provided by the designers as a language for dramatic improvisation.

Whether the player has a strong sense of plot or not, their (inter-)actions are likely to have consequences not only directly within their own play experience, but also implicitly, implying a game world beyond that which is explicitly represented to the player. In pure narrative systems this is the *story* level. In computer games, the temporal system may not be as precisely pre-structured as traditional narratives, and so can more appropriately be referred to as the *simulation* level²; this is the level at which the authored logic and parameters of a game system together with the specific interactive choices of the player determine an (implied) diegetic (i.e. represented) world, only some of which is made available to the player via the experiential zone created by a virtual camera, a virtual volume of audio reception, and a surface of virtual haptic reception (e.g. a virtual body that receives damage or health). For example, in playing *Sim City* (Infogrames, 1989), the screen reveals to the player a small section of a city that is a part of the much larger implied world of the total city that is simulated on the computer and heavily determined by the players interaction (establishing zones, infrastructure, etc.).

Beneath the simulation level is the level of the *generative substrate*, the system of functions, rules and constraints constituting a space of possible worlds of experience created by the designers of the game. In the traditional semiotics of verbal and written language these distinctions may be equated in terms of the discourse level corresponding with speech (the semiotician Saussure's *parole*) and the generative substrate corresponding with the system of a language (Saussure's *la langue*; see Stam et al, 1992). Narrative theory posits semantic levels including the diegesis or story (the world and its causally interrelated events represented by discursive acts), the plot as an expressively modulated selection of the events taking place within the diegetic world, and the discursive order of presentation of the contents of the plot (the narration or discourse level). Since ludic systems may have time orders that are not dominated by strong narrative models, a different terminology is required to characterize these various levels. Hence it is possible to distinguish the corresponding and analogous levels described above: the discourse level, the performance level, the simulation level and the generative substrate. These correspondences are depicted on Figure 1. Note that ludic systems are significantly different at the level of the generative substrate. For verbal language and narrative, the structural substrate is understood as a space of possibilities implicit within a culture and from which members of the culture may improvise meaningful stories (e.g. as described in Propp's, 1968, analysis of Russian folk tales). Computer games may be understood as deriving from a similar space of culturally implicit possibilities. But for computer games there is a narrower and much more specific generative basis, derived from general cultural understandings but embodied in the software code of the game framework, thereby supporting a very particular space of possible game worlds that may be created at the simulation level in response to player interaction. Hence for games in which computers play a significant role in creating the game world, the generative or structural substrate can be

² Here simulation is understood broadly to include fabrication of fictional scenarios.

divided into two levels, one inspiring the code and being very similar to the structural foundations of narrative systems, and the other being the actual structure of the code as an artifact.

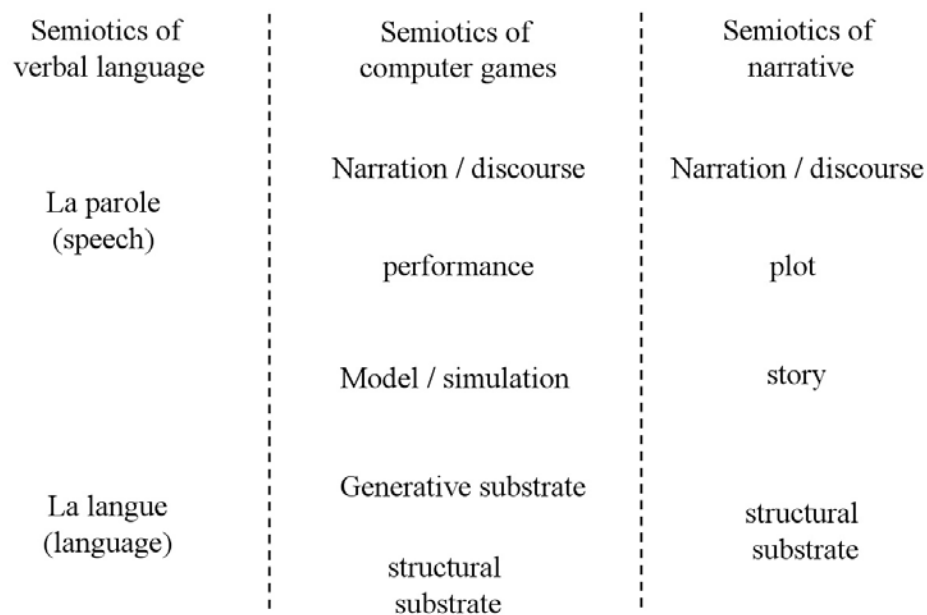


Figure 1. Approximate correspondences of layers of the communication systems of natural language, ludic systems and linear narratives.

For a player to experience different time structures at the performance level, specific design and implementation methods are required at the simulation and generative levels of a ludic system. Here we are concerned with the internal structure of temporal development, a different (but compatible) emphasis to the temporal genre distinctions proposed by Aarseth et al (2003), who focus on pragmatic and discursive issues involving distinctions between mimetic versus arbitrary narration/presentation times, real time versus turn-based interaction cycles, and the discourse-to-simulation level issue of overall finite versus infinite persistence of the ludic space. Similarly, the formulation of game semiotics presented by Klabbers (2003) also appears to be compatible with the formulation presented here. However, Klabbers emphasizes atemporal aspects of game semiotics (their *synchronic* semiotics) in relation to simulation, while the framework presented here emphasizes temporal (or *diachronic*) semiotics in relation both to simulation, game and narrative time structuring principles.

Game, Simulation and Narrative as Independent Formal Subsystems of Ludic Space

As noted above, ludic systems can be analysed in terms of several layers of semiotic encoding, including a simulation layer, a game layer and a discourse layer. While these layers represent semiotic sublevels of the surface text of a ludic system constituting the played experience (the history of the screen, in the case of a computer game), they also represent potentially independent semiotic domains having their own design principles and traditions. Choosing to design at a specific level is to adopt the design conventions and features appropriate to that level, with specific corresponding affects for the player.

Simulation

The simulation level represents the lowest level of temporal design concern, addressing features such as the basic realization of motion and other functional characteristics of a world; in a computer-based system this amounts to the design of potential changes in the state of the ludic system from frame to frame, i.e. the design of what happens with each simulation cycle or tick. This level is not limited to computer games; table-top games, such as role-playing games and war games, frequently involve a convention for what the time step involved in a game turn represents in diegetic time, while live-action role-playing games may adopt similar conventions or depend upon physical performance to fabricate this level of the game story.

Much has been made over the last few years of the view of games as simulations (e.g. Frasca, 2001). But what is a simulation in general as distinct from a narrative or a game? A simulation can be defined as: *a representation of the function, operation or features of one process or system through the use of another*. Simulation may be the dominant semiotic domain of some ludic systems, such as flight simulators. In this case the ludic system may involve no specific game play or narrative patterns. The time patterns that emerge over the course of running a simulation may be completely different for different runs of the simulator and may never have been anticipated by its designers. Repetitive action may be used to operate a simulation model, but this may not be directed to any overall goal predefined for the player/operator by the system's designers. A good example of this is *Sim City* which is based upon a cellular automaton model in which the state of any zoned unit on the world surface at a particular simulation tick is a function of its type and both its own state and the state of its immediate neighbours in the previous simulation tick; the player may take the city in any direction of development at all, with no particular end point.

Designing the simulation level of a game can draw upon many techniques from discrete simulation of continuous systems, including methods for designing quantisation in both time (how many discrete steps a continuous time interval is subdivided into, and the sample time for each step) and space (how many bits represent a continuous spatial extent). Quantisation effects can have a direct bearing upon the player's perception of what happens within a virtual game world. For example, a walk cycle must be represented with more than two simulation ticks per cycle, or else the motion may appear static, occur at an incorrect rate or go in the wrong direction (these being *aliasing* effects). Discrete event simulation offers techniques for modeling systems as networks interconnecting stochastic arrival rates with services having stochastic service times via queues with various properties (first-in-first-out, last-in-first-out, random service order, queuing behaviors such as jockeying and reneging, etc.).

The richness of a ludic system as a simulation is reflected in the ability of players to define their own stories, experiences, games or game systems within the simulated world. In this sense a simulation provides a *field of play*, within which it is up to the players to define how they will play.

Games, Play and Play Time Structure

While there is a lot of ongoing work aiming at defining games, a game will be defined here in a rather narrow way as: *a goal-directed and competitive activity conducted within a framework of agreed rules*.

This can be referred to as the ludic or ludological definition of a game, the kind of definition at the base of traditional mathematical game theory. This definition implicitly or explicitly

captures many features of the definition developed by Juul (2003), in particular encompassing Juul's first three defining characteristics of games: rules, quantifiable outcomes and values assigned to those outcomes. The definition does not include Juul's additional criteria of player effort, player attachment to outcomes, or negotiable (real-life) consequences. These latter criteria are issues of pragmatics that are independent of the internal formal system of the game and highly subject to external accidents of history and context. Our definition leads to the possibility of recognizing the formal system of a game at work where the players may not self-consciously regard their activity as playing a game (including, for example, the reference systems with themes noted by Klabbers, 2003, including business and public administration, health care, military operations and religion).

Game Rules

The rules of a game establish what as a player you can or cannot legally do within the game and what the behavioral consequences of actions may be within the game. Successful play does not necessarily require learning all of the game rules, especially in the case of computer games where a player learns how to interact to make progress within the game within the rule-bound space of possible interactions enforced by the computer system.

Time Structures of Game Play

The time structures involved in game play include game moves, game-play gestalts (or low level player interaction patterns), tactics and strategy, and larger scale organizations of game and play sessions.

A *move* within a game is an abstraction over player action, mapping action to a specific significance within the rule set independently of local, personal and idiosyncratic variations in performance; a move is a connotation of a physical or simulated action allowed and facilitated by the framing of the game (I can move a chess piece on the board at any time, but I only make a move in the game of chess when I'm playing the game). Hence a player performs actions having conventional connotations as moves within the formal system of the game. Those actions are likely to be highly stylized according to the game, and actions too dissimilar to the stylized set will be regarded as fouls or cheats if their performer intends them to have in-game significance, or as *extra-ludic* actions potentially frustrating other players if they are not intended to have in-game significance. Moves generally represent a larger scale logical structure on top of the physical or virtual realization/fabrication of the game world (i.e. the simulation level); e.g. in chess 'rook to f5' is a move that may be realized instantaneously if the game is played abstractly by text (in which case the game world only exists imaginatively), by an endless range of different possible physical movements in the play of a board game, or by the simulated glide of a chess piece, realized frame by frame, in a computer game. The abstract nature of moves is reflected in combat interaction in games like *Dark Age of Camelot* (Mythic Entertainment, 2003), where the player may 'select a weapon', 'target an opponent' and 'initiate combat', while the detailed realization of combat moves is automated. For some other games, like *Morrowind*, the player has a more interactive role in determining how combat is executed, including the constant need to orient the character towards the target and to initiate each strike or move of combat interaction, thereby having a greater role in the implementation of game moves as a performance, including the basic frame-by-frame control of the character's position and orientation at the simulation level.

Game-play Gestalts As Patterns of Moves

Learning to play a game, making progress within a game and completing or winning a game are matters of learning how to interact within the game system and its rules in a way that supports progress. This is a matter, not necessarily of learning the game rules, but of learning a *game-play gestalt*³, understood as a *pattern of moves* within the game system. Playing the game is then a matter of performing the gestalt. It is what the player does, within the system and as allowed by the rules of the game. In computer games, where the machine enforces the rules, this may lead to players having very poor conscious appreciation of what any of the rules actually are; instead they have learned successful (and unsuccessful) patterns of interaction by trial and error.

A game-play gestalt can have many forms for a particular game, capturing different playing styles, tactics and approaches to progressing through the game and (perhaps) eventually winning. In general, it is a particular way of thinking about or understanding the game state from the perspective of a player, together with a pattern of repetitive perceptual, cognitive and motor operations. A particular game-play gestalt could be unique to a person, a game session, or even a playing session. Recurrent game-play gestalts can also be identified across games, game genres and players. Some examples of game-play gestalts in computer games include:

Action games: shoot while being hit, strafe to hiding spot, take health, repeat

RPGs: send fast character to lure enemy from group, all characters kill enemy, take health, repeat

Strategy Games: order peasants, send to work, order soldiers, send to perimeters, repeat while slowly expanding the perimeters (up to the point of catastrophic win/lose); OR: move x archers to tower y every n minutes to head off the enemy camel musketeers from the east who arrive every n minutes

In General: confront barrier, save if barrier overcome, reload and retry if unsuccessful

These kinds of patterns may or may not be explicitly designed for by the creators of a game. They are play patterns rather than game design patterns of the kind identified by Björk and Holopainen (2005). If designers do take them into account, it may be in supporting the development and emergence of these patterns in play, rarely by forcing them on the player.

Tactics can be regarded as larger scale and conscious game play patterns developed by players. Hence there is a progression of scale in play patterns. Moves are primitive meaningful game actions, performed within the game space, using relevant game objects and conforming to the rules of the game. In turn-based games, the completion of a move (or a specific number of moves required or allowed by the game rules) generally signals the end of a particular player's turn. A repeated pattern of moves developed by a player as a method of making progress within the game is a game-play gestalt. Tactics can be defined as game-play gestalts or patterns of game-play gestalts that are consciously chosen by a player in response to the actions of an opponent or other aspects of a developing situation. Tactics are selected with the aim of winning bouts or rounds of a game.

³ A *gestalt* may be understood as a configuration or pattern of elements so unified as a whole that it cannot be described merely as a sum of its parts.

A strategy can then be understood as a larger scale policy or plan, consciously followed by a player in order to win at the end of a larger scale time structure of game sessions, such as a match, a contest, a league or a tournament. The most primitive complete game experiences, at which a point of win or lose is reached, are bouts or rounds. Larger scale formal game structures tend to be highly repetitive patterns of these simple game experiences. They are mostly concerned with the organization of opponents, extending the simple competitive situation of a game to include a broader field of opponents with a view to obtaining a global performance or game play ranking by accumulation of the results of many bouts. Large scale game structures have their own rules for accretion of the results of bouts and rules for matching competitors in ongoing events. A multi-game structure requires a *principle of accrual of results*. That is, various formulae may be used for accumulating wins and losses, and degrees of win/loss, into an overall competitive ranking, or for the identification of a set of champions across various categories. The structure may also include *elimination events* in which losing competitors are eliminated from further competition, or the game system may include *principles of handicap* by which differences in demonstrated game play expertise are compensated for to provide for less predictable outcomes in ongoing competitions.

The time structure among these larger scale game groupings is incidental to the essential performance of the players. Even more strongly, it can be stated that the large scale structures of game forms have little to no dependence on specific time orders. Their primary meaning is the ranking of player competence; time-ordered competitions are a convenience for identifying this ranking. In principle it doesn't matter at all what the sequencing of competitions is, as long as it leads to an order of player competence (hence the common freedom to choose the sequence in which one defeats one's opponents in a computer game). This is a critical distinction between the temporal form of games and narrative patterns having a strong a priori linear time structure (or set of potential linear time structures) created by their designer. In computer games, instances of combat are individual bouts (these are experiences of playing single games, by the definition above), while game levels are organized as a series of matches, contests, leagues or tournaments. If a larger scale game structure is designed to present players with a specific sequence of game experiences, activities and opponents, serving to shape the emotional tone and intensity of the experience, the form is starting to move away from pure game form, more strongly integrating variants of authored narrative as manifested in the pre-specified sequential design.

Narrative

As noted above, a narrative is a representation of the causally interconnected events of a story. The structure of this representation will generally conform to one of a variety of specific structural patterns. The structure of narrative has its own specific conventions and design traditions, as exemplified by the work of Vogler (1998).

A very common narrative structure used in computer games, borrowed from film scriptwriting, is the *three-act restorative structure* (Dancyger and Rush, 1995). The three-act restorative structure has a beginning (the first act) in which a conflict is established, followed by the playing out of the implications of the conflict (the second act), and is completed by the final resolution of the conflict (the third act). The three-act restorative structure includes a central protagonist, a conflict involving a dilemma of normative morality, a second act propelled by the hero's false resolution of this dilemma, and a third act in which the dilemma is resolved once and for all by an act that reaffirms normative morality. Each act within the

three-act structure culminates in a point of crisis, the resolution of which propels the plot into the following act, or to the final resolution.

In computer games that use the three act restorative structure as a strongly imposed framing structure (e.g. *Quake*, id Software, 1996, and *Dungeon Siege*, Gas Powered Games, 2002), the central conflict form often manifests recursively (i.e. the structure is repeated at different temporal scales). For example, the overall restorative three-act model may be applied to the game experience as a whole, with the dramatic arch being completed when the user finishes the game. At this scale the story is usually not interactive, since act one, key scenes within the story of act two (i.e. primary plot points), and the playing out of the consequences of the final resolution in act three are typically achieved by cut scenes, sequences of non-interactive, pre-rendered video or non-interactive animation sequences. The next scale down within the recursive structure is that of the game level. The game level is designed for the pursuit of a goal, that of the player reaching the end of the level, which progresses the player through the second act of the larger scale three-act structure of the game narrative. There is rarely if ever a one-to-one correspondence between game levels and acts; more typically, the first act and the end of the third act are presented via cut scenes, with playable game levels summing to form a highly extended second act followed by the final resolution of the third act as the end of game play (e.g. by overcoming the final and toughest enemy, usually a demonic character at the heart of the central conflict in the story). Although experience within a game level typically has much of the structure of a match, a contest, a league or a tournament, the sense of narrative development can be enhanced by increasing difficulty through a game level, or by an internal dramatic structure that emphasizes the point of completing the game level, such as the defeat of a level boss, the barrier creature at the end of the game level. The false resolution that drives act two of the three-act restorative model at the highest structural scale may be seen manifesting repetitively with each game level: when the game level is resolved (completed), the player finds themselves at the beginning of the next game level full of conflicts.

There are many strategies by which story and narrative elements may be integrated with detailed game play. Space prevents elaborating this in detail here, although the issue is dealt with at length in Lindley (2005), where it is proposed that player reactions and responses to different strategies and the play experiences resulting from them have as much to do with player tastes, styles and motivations as they have to do with the overall strategy adopted.

Whatever the overall approach to narrative adopted by a game's designers, it is the performance of game moves that consumes most of a player's cognitive resources. Game moves provide a version of what Mackay (2001) refers to as *fictive blocks*, basic fragments or units of fictional/narrative significance that may be strung together to form a larger scale narrative. Mackay takes fictive blocks divorced from their original context to be equivalent to Schechner's *strips of imaginary behavior*, patterns that constitute a repertoire of potential behaviours that are performed by an actor in new arrangements in ways that may appear spontaneous and unrehearsed. Fictive blocks derived from popular culture sources (films, television, literature, etc.) are understood to circulate broadly within a culture, where they are available for reappropriation by its participants for the creation of new narratives (novels, movies, role playing game sessions, etc.). Although not considered by Mackay, in the case of a computer game fictive blocks have a tangible and predefined form created by the game authors as the constrained set of valid game moves that the player may choose from at any particular point in the unfolding play experience.

From a narrative perspective, game moves can therefore be regarded as the basic language of the player's narrative performance. Game play is often regarded from the perspective of a series of challenges (e.g. Rollings and Adams, 2003), and this is implicit in the above definition of a game. This has an interesting analogy to the use of inter-character conflict to generate dramatic interest in the performance of cinematic or theatrical scripts. As Lindley (2005) argues, immersive or performative dramatic game play therefore requires a confluence of the basic performance primitives provided to the player as moves with any larger scale story or narrative structures of a game or ludic system. The design of the performance primitives provides an orientation towards their dramatic significance. Highly repetitive challenges leading to the repetitive performance and refinement of a game-play gestalt orients the focus of dramatic energy towards the outcomes of small scale game structures (each challenge is a bout of the game) and away from any larger scale sense of story or narrative. The drama of play is then the drama of the game for the player, strongly tied to the outcomes of each small scale game challenge in a sequence of very many of those challenges (e.g. hundreds of enemies to be defeated in turn). In this case, when a narrative frame is presented via cut scenes, it functions as a break from immersion in small scale game play, which may be effective if used at the right time within the rhythmic structure of performance difficulty designed into a game. However, this focus upon the game bout involves a focus upon the player rather than their character, and any larger scale narrative framing delivered by cut scenes may be hard to follow since it has so little to do with the details of play.

A strong sense of dramatic characterisation requires ongoing character development at the interactive focus of play. Repetitive game-oriented play restates the character identity of 'I am a player', with a focus upon achievement, e.g. in the form of experience points and character levels. Play oriented towards characterization requires the moves of the game to be geared towards answering the question 'who am I?' as a character within the game world. The player's answer to this question of identity, as expressed in their choice of character moves, must also be synchronized with being 'told who you are' in any non-interactive narrative material such as cut scenes.

These modalities of play can be well, if anecdotally, illustrated by the game *Morrowind*, which provides for a wide range of different ways for the player to define the identity of their character. *Morrowind* can be played with a highly game-oriented focus, simply killing enemies and accumulating wealth, skill points and experience points with a strong sense of player progress but a very weak sense of who the character is. However, *Morrowind* also provides several mechanisms for defining a social, affective and instrumental identity for the character. These mechanisms include one of ten 'races' (dark elf, nord, khajit, argonian etc.), any number out of twenty one professional classes to join and develop (e.g. assassin, crusader, battlemage), twenty seven skills that may be advanced (e.g. athletics, medium armor, conjuration), eight basic abilities functioning as prerequisites for skill development (agility, endurance, intelligence, luck, personality, speed, strength and will power), and any of four 'great houses' to join. The social categories of race, class and house provide social identities within the game that can lead to many possible situations of conflict, and the actions of the character have a bearing both upon skills developed and the deepening or not of the character's (potentially many) virtual social allegiances. These distinctions result in a dense network of in-game social relationships and roles constituting a comparatively rich social and dramatic identity for the character. A player character begins the game as an 'outlander', a foreigner of unknown background on an unknown mission. Game moves oriented towards character development have the form of agreements and social roles presented by non-player characters (NPCs) via dialog boxes, with responses chosen by the player. Progress within

social categories is in many cases achieved by the completion of ‘duties’ or quests agreed to via these game moves and often requiring the character to overcome many forms of game challenges of the more repetitive game-oriented kind (e.g. combat). Significantly, the meaning of the game moves constituting choices within dialog boxes is determined by the text content of choices, and that content is not highly repetitive. With each choice an identity may be adopted, a role or task undertaken, and with that identity or role, and with each task completed or not, the character gains a partial answer to the question ‘who am I?’. This search for identity has its strongest manifestation if the player chooses to follow the central plot line of the game, in which case the game becomes a search for the answer to the central question of identity: ‘am I the hero Nerevar, prophecied reincarnate of the ancient hero?’.

Choosing to play the game with this focus on characterization and identity is to follow classical dramatic paths of character development. In this case repetitive combat-oriented play becomes a secondary consideration after identity formation, and the highly repetitive and interactively undemanding moves required for dealing with dialog systems fall into the background in relation to the more unique text content of the dialogs and the choices made for articulating character. The lines of text offered to the player as dialog responses by their character add up, in a game like *Morrowind*, to a very large diversity of unique fictive blocks that the player may use to perform their character. This is perhaps not surprising when it is considered that table-top and live action role-playing games, both generally much more effective for character development than computer games, rely heavily upon verbal performance. An ongoing challenge for computer game design is to achieve a strong sense of dramatic play as a game character using games moves that do not rely so heavily upon text.

A Unified Classification Plane

The three kinds of temporal semiotic systems found in ludic systems, simulation, the ludic game and narrative, might be found together or separately in any specific system. Among ludic systems in which more than one of these forms is present, there may be variations in which system provides the primary modality in structuring the player’s interactive engagement with and experience of the system. This is reflected in concrete terms by the range of different player-selectable actions belonging to each of these kinds of formal systems offered by a specific design. Taking *Morrowind* as an example, it might be said that simulation-based actions include character movement within the world, positioning and management of game items (weapons, clothing, etc.), and creating spells, potions and enchantments. Game-based actions include actions associated with meeting the kinds of challenges noted by Rollings and Adams (2003), including actions for operating the combat system of the game (combat being fundamentally a game by the definition of game used here). Narrative and story-oriented actions then include choosing to belong to social groups (races, classes, guilds and houses), accepting and completing quests, engaging in NPC conversations and choosing to allow cut scenes to play. This category would also include the kinds of inter-player communicative actions identified by Manninen (2003) in the case of multiplayer games. A specific design can then be described in terms of the availability of player actions falling into these three categories, and the intended distribution of these action categories within the overall play experience.

To illustrate this, it is possible to construct a classification plane as a triangle with one semiotic form at each point, as shown on Figure 2. It is then possible, as a rough heuristic for comparing different games and genres, to place games and genres on that plane, emphasizing the relative degree to which the different levels of temporal structure are the intended focus of

player engagement. Of course this is only a rough heuristic, and reflects the balance of intended engagement modalities embodied in a design by the system designers; for any particular design focus, players may choose or attempt to interact in ways reflecting a different balance of forms.

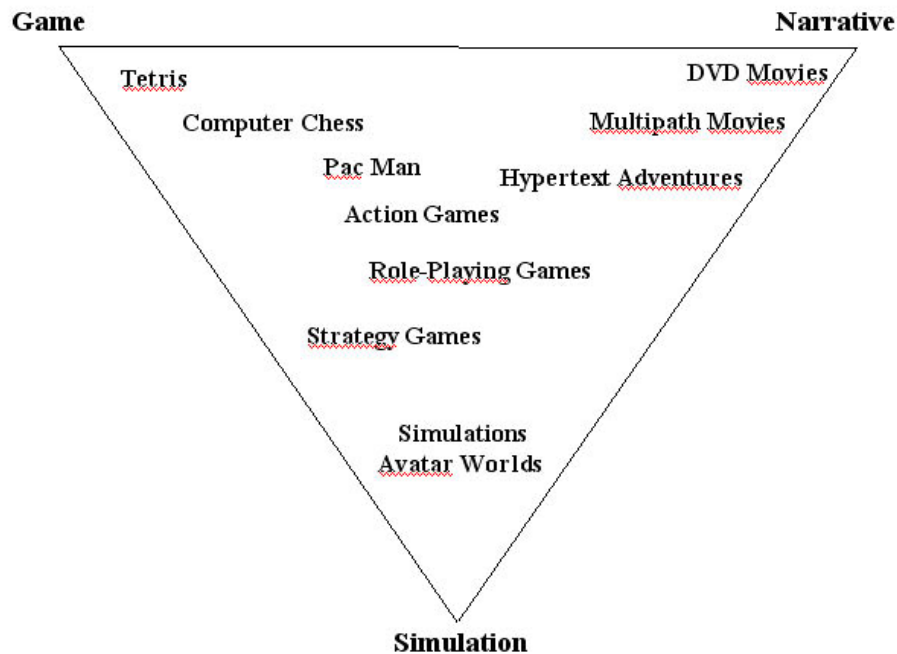


Figure 2. A classification plane based upon time form in ludic systems.

This can be illustrated by considering several examples. Avatar worlds and vehicle simulators are placed at the simulation extreme. Early avatar worlds were interactive three-dimensional virtual spaces in which a user is represented by a movable avatar (see Damer et al, 1999). These worlds rarely presented much to do, however, since they lacked any game or narrative content. The design intention is to provide freedom for the players to interact in ways typical of the basic simulated 3D world, generally concentrated upon movement within that world. The design does not provide any pre-designed game structures or story elements, although players could improvise these within the limits of freedom available within the world. Here the actions available to players are simulation-oriented actions pertaining to movement, together with inter-player chat.

Board games and games that do not represent any kind of fictional world, such as Tetris, belong at the game play extreme. These games are very abstract, but still engaging. There is a simulated game space upon which basic game play is founded, but the designed intention is for players to play the game, not to explore the underlying simulation in any other way, this in fact being impossible in many games (e.g. *Space Invaders*, Taito, 1978, or *Tetris*). In this case the only player actions available are game moves.

At the narrative extreme are the fixed narrative structures of digital linear movies. Multipath movies (e.g. <http://www.brilliantdigital.com/solutions/movies/>) present actions expressing choices for the viewer at particular branching points within the film story; there is no simulation level (creating a world is addressed by the internal semiotics of the film segments) and no built in game play in terms of player actions representing game moves, although players could invent a game, e.g. 'find the longest/shortest path to an end of the movie'.

Action games, strategy games and RPGs incorporate prominent features of all forms, being games, simulations and often having significant high level narrative structure. This can result in the kind of variability of play described above for *Morrowind*. Differences in the dominant semiotic modes here are reflected in, for example, the stronger emphasis of role-playing games upon story, as expressed by their greater number of options for story-oriented player actions, such as accepting quests and joining social groups.

Conclusion

This paper has presented a framework for ludic semiotics based upon the recognition of the different temporal semiotic systems of the simulation, the game and the narrative, each of which has its own design principles and modes of player engagement. It has been proposed here that different modes of player engagement within these different semiotic systems are facilitated by the availability of different kinds of game actions. This association of actions with semiotic systems is a topic of ongoing investigation and open to interpretation, especially since some actions may function within more than one semiotic system (e.g. a combat action is both a game move and a story performance primitive). Ongoing work in this area is particularly concerned with validating these associations, especially in the context of player affects as they may be related to different player motivations, play styles and other cognitive factors.

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